

Intelligent Pipetting System Towards Automatic Liquid Handling Applications

Lalitha Pratyusha Bheemavarapu
Dept. of Electrical Engineering
Indian Institute of Technology, Madras
Chennai, India
ee16s005@ee.iitm.ac.in

Malay Ilesh Shah, Ravishankar Ramanathan
Healthcare Technology Innovation Centre
Indian Institute of Technology, Madras
Chennai, India

Mohanasankar Sivaprakasam
Indian Institute of Technology, Madras and
Healthcare Technology Innovation Centre
Chennai, India

Abstract—Automation of life sciences and pharmaceutical laboratories is high on demand in the present day, especially with the emergence of high throughput screening and drug discovery. Lab automation provides reliable and efficient solutions, limiting reagent amounts with increased throughput and reduced costs. In this work, we present an affordable, intelligent, sub-microlitre resolution automatic pipetting system. This system employs a differential pressure sensor to detect most frequently occurring pipetting errors in automatic liquid handling workstations, which include no liquid sample present in the source well, pipette tip not attached to the holder, clogging of the tip, aspiration of air bubbles or foam, insufficient liquid in the source well, improper dispensing of the samples and tip not intact with the holder. This also determines liquid volumes aspirated into the pipette tip, providing a self-validating mechanism for the system, thereby reducing human intervention and improving performance efficiency. A prototype system was developed in-house and gravimetric analysis was carried out to determine system performance. Repeatability coefficient of variation for the designed system was found to be below 5.5% for the volume range of 10 to 200 microlitres, while the error obtained was observed to be less than 1.6%. Experimental tests conducted could flawlessly detect mentioned pipetting abnormalities through the algorithm designed.

Keywords—automation; liquid handling; intelligent pipetting; calibration free; affordable; non-contact dispensing; life sciences

I. INTRODUCTION

The primary objective of a liquid handling system is to transfer desired volumes of liquid samples between different receptacles, without any cross contaminations or errors. Various experiments like gene sequencing, drug discovery and screening, protein crystallization, assay processing, high throughput screening performed in life science and pharmaceutical laboratories involve handling thousands of liquid samples of micro and nanolitre volumes, with a demand for a high degree of accuracy and precision [1], [2].

Manually operated mechanical pipette forms a basic liquid handling device, wherein performance efficiency of the system

is operator dependent and needs manual setup each time a liquid volume has to be varied. It becomes time-consuming and highly impossible to handle thousands of biosamples with these handheld tools [3]. The performance consistency of these systems largely depends upon the individual handling the pipette and also varies from person to person [4]. These are minimized with automated pipettes, wherein the system performance is user independent. But, these systems still depend on manual methods for positioning and transferring of samples. Robotic liquid handling systems further increase the level of automation by integrating automatic pipetting systems with positioning systems to transfer samples between the source and destination wells. They bring down the level of human interference and thus provide consistent fatigue-free performance with good accuracy and repeatability. Such robotic systems being more advantageous over the manual methods find huge applications in biological, chemical and pharmaceutical laboratories, but tend to be expensive [1], [5].

Automatic liquid handling systems available in market vary from a simple sample transfer device to complex integrated workstations. There arises a conflict between cost, size and level of automation in the existing systems. More the demand for automation in these systems, higher would be the cost. Recently introduced Opentrons liquid handling robot is one low cost system available in the market, which uses a manual pipette integrated with 3-Axis robotic platform [6]. Though affordable, this system doesn't have any inspection over the liquid handling process to verify for proper positioning and pipetting. This forms a major setback of such systems leading to a high demand for intelligent and affordable liquid handling systems, which integrate sensing and feedback control mechanisms for achieving accurate and precise handling of samples [1].

In this paper, an intelligent automatic pipetting system that handles liquid volumes ranging from 10 to 200 μ L, with sub-microlitre resolution was designed. It is integrated with a pressure sensor, which monitors the whole pipetting process and detects irregularities if any, including the most common

errors encountered in automatic liquid handling workstations. Pressure sensor employed also measures the volume of liquid samples aspirated into the tip, which is a major feedback check for the operator. The designed system integrated with 3-Axis robotic platform provides an affordable intelligent solution for automation in the field of liquid handling.

Hardware design of the pipetting system, sensor and system integration are described in Section II. Section III provides a detailed explanation of algorithm devised for detecting aforementioned irregularities. Experimental procedures carried out for determining accuracy and precision of the system are presented in Section IV. This section also deals with liquid volume measurement in the pipette tip. Section V provides an understanding of the experimental results obtained, with conclusion remarks made in Section VI.

II. SYSTEM DESIGN

Basic block diagram of the designed system is as shown in Fig.1, indicating three main components of the system. For initial prototyping purposes, National Instruments modules were used. Software panel provides a user interface for control of the pipetting process. Electronic interface unit bridges software control and the electromechanical pipetting system. This also involves the acquisition of pressure data from the sensor, data acquired is fed back to the software panel for processing.

A. Hardware Design

The designed pipetting system uses the principle of air channel displacement for handling liquid volumes, wherein a dead air column present above the pipetting tip is displaced up and down for aspiration and dispensing of samples respectively. This is achieved by the movement of a piston within an airtight cylindrical sleeve. Fig. 2 illustrates the basic design of pipetting system. Upward movement of piston reduces pressure inside the system creating a vacuum, thereby pushing liquid to be aspirated into the pipette tip. The volume of liquid aspirated is equal to the volume of air column displaced. Similarly, the piston is moved downwards for dispensing liquid volumes.

Mechanical design of pipette was done in-house using 3D design software. Pipette body and piston were printed using Fused Deposition Modelling (FDM), while Multi-jet printing was used for printing airtight sleeve. Tip holder of the system is designed to support standard 200uL disposable pipette tips. For vertical displacements of the piston, it is mechanically coupled with a unipolar, stepper motor 26DBM05D2U-L

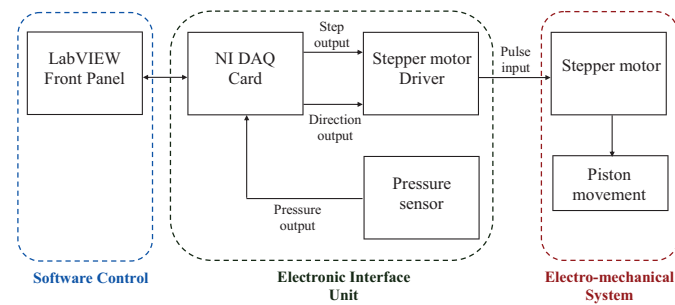


Fig.1. Basic block diagram of the prototyping system

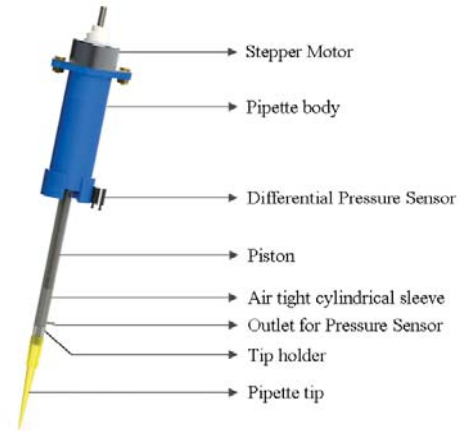


Fig.2. Hardware setup of pipette indicating different parts of the system

(Portescap, USA), having a 0.5° step angle and 12.7 micrometer linear pitch. With the current hardware setup, it provides a volume change of 0.22 uL per full step. Motor is configured at 256 micro stepping and hence the volume change for each micro-step is further reduced by a factor of 256. A stepper driver is used for controlling motor displacements.

B. Sensor Integration

During pipetting process, pressure changes in the dead air column present between the piston and the pipette tip are acquired to verify if liquid samples are properly aspirated and dispensed. A slot provided for the same on the airtight sleeve of the designed system can be observed from Fig. 2. This slot is a small tubular extension from the sleeve which can be connected to the pressure sensor placed on pipette body through an airtight rubber tube.

An analog voltage output type differential pressure sensor HSCDRRN010MDAA3 (Honeywell International Inc., USA) of ± 10 mbar pressure range was used. This sensor has two input ports and provides an output proportional to the difference between pressures applied to each port (Port1–Port2). In our application, Port 1 of the sensor is placed in the air column whereas Port 2 is exposed to the atmosphere. The resulting pressure (P_{out} in mbar) would be the difference of pressure in the air column and atmospheric pressure. As shown in Fig.1, an NI DAQ card is used to acquire data from the pressure sensor.

C. Electronic Interface and System Integration

A USB-6363 Data Acquisition Card (National Instruments Corporation, USA) and a TMC261-PA (Trinamic, Germany) stepper driver are used to control stepper motor, which in turn regulates the piston movement. Two main mechanical controls of a pipette include the direction of piston movement, which determines whether the liquid sample is to be aspirated or dispensed and the amount by which piston is to be displaced to control the volume of liquid being aspirated or dispensed. Stepper driver takes the direction and step inputs from the digital and counter output pins of DAQ respectively and provides corresponding pulses to the stepper motor, thereby accomplishing the desired pipetting operation. The prototype system developed and hardware setup are indicated in Fig.3.

Pressure information from the sensor is acquired using analog input channel of the same USB-6363 Data Acquisition Card. Sampling rate of 1000 Hz is set for the complete data acquisition process.

The user interface for the pipette control is developed in LabVIEW (National Instruments Corporation, USA), Fig.3(c) depicts front panel of the virtual instrument designed. Data transfer between the DAQ and software panel is done over USB communication. User controls involve providing desired volumes of pipetting samples and if the samples are to be aspirated or dispensed. Once these inputs are provided and run, DAQ provides corresponding pulse and direction signals to stepper driver, which control the motor displacements in turn. The same virtual instrument is designed to handle pressure data acquisition as well. Data acquired is processed and validated for any irregularities and operator is alerted on the same.

Proposed system design would integrate above mentioned controls into a single microcontroller based system, where few buttons and displays can be incorporated for providing a user interface.

III. PRESSURE DATA ACQUISITION ALGORITHM FOR INTELLIGENT PIPETTING

An algorithm was designed to capture pressure changes during pipetting and process the data acquired to evaluate for proper aspiration or dispensing. Single pressure sensor used in this setup with the algorithm designed detects below-mentioned checks, thereby facilitating precise handling of liquid volumes.

- No liquid aspirated into pipette tip
- Presence of air bubbles or insufficient liquid in the source well
- Clogging of the tip
- Leakage due to tip misalignment
- Liquid level in the pipette tip
- Droplets hanging from the pipette tip due to improper dispensing

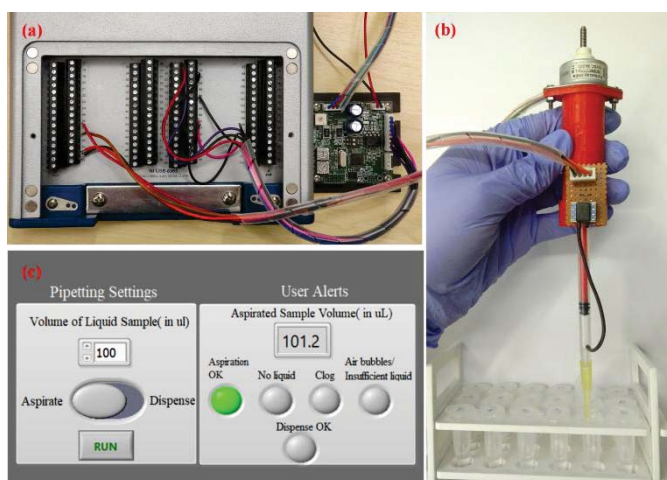


Fig. 3. (a) Hardware setup of the prototype system. (b) Liquid samples being aspirated with the designed system. (c) LabVIEW front panel of the pipetting control VI.

A. Aspiration Mode

In case of a normal aspiration, pressure can be observed to decrease as liquid volumes are drawn into the pipette tip. Output of the pressure sensor which is difference of the atmospheric pressure from pressure in the air column can be observed from Fig. 4(a). This curve is obtained for aspiration of 200 μ L distilled water. Once the liquid is completely aspirated, pressure in dead air column remains constant and is dependent on the height of the liquid column in the tip. With the current provided settings, aspiration time varies from around 0.1sec for 10 μ L to 2.5 sec for 200 μ L. Pressure data is acquired for additional 2 sec after the aspiration is done.

When a first derivative is taken over the obtained pressure curve, it is observed to have one sharp valley and peak as indicated in Fig. 4(b). Valley detected corresponds to a sudden decrease in measured pressure from the atmospheric pressure at the beginning of aspiration, as the tip is now exposed to the sample being aspirated. Similarly, once the aspiration is done, the pressure is observed to attain a constant level, resulting in a sharp peak. This characteristic of the ideal aspiration curve is used for determining few of the irregularities mentioned, as shown in Fig. 4. An Algorithm designed for validating different non ideal conditions during aspiration mode are as explained.

1) No liquid aspirated into pipette tip

This condition often occurs when there is no liquid in the source well or no tip attached to the tip holder or incorrect positioning of the pipetting arm in liquid handling workstation. Pressure waveform attained for all the three conditions is as depicted in Fig. 4(b). Once the pipetting starts, detection of these conditions is done by calculating slope after 5 ms of acquisition. Slope when liquid is present was practically observed to be in orders of less than -0.1 and for no liquid aspirated to be greater than -0.01. With the set sampling rate, these nonidealities can be detected within 5 ms from the start of pipetting and the operator is alerted on the same.

2) Presence of air bubbles or insufficient liquid in the source well

Pressure curve observed for these nonideal conditions is depicted in Fig.4(c). These abnormalities are validated by detecting the number of peaks and valleys in the corresponding first derivative curves. An appropriate threshold has been applied to eliminate insignificant small amplitude peaks present throughout. It can be observed from Fig.4(c) that these result in more than one peak and valley. The operator is alerted of these irregularities over GUI, once the number of detected peaks and valleys are greater than one each.

3) Clogging of pipette tip

Fig. 4(d) shows the pressure curve attained and the corresponding first derivative curve when a pipette tip gets clogged. Number of peaks and valleys are detected and compared with ideal values. When pipette tip gets clogged, there would not be any peaks, only valleys can be observed.

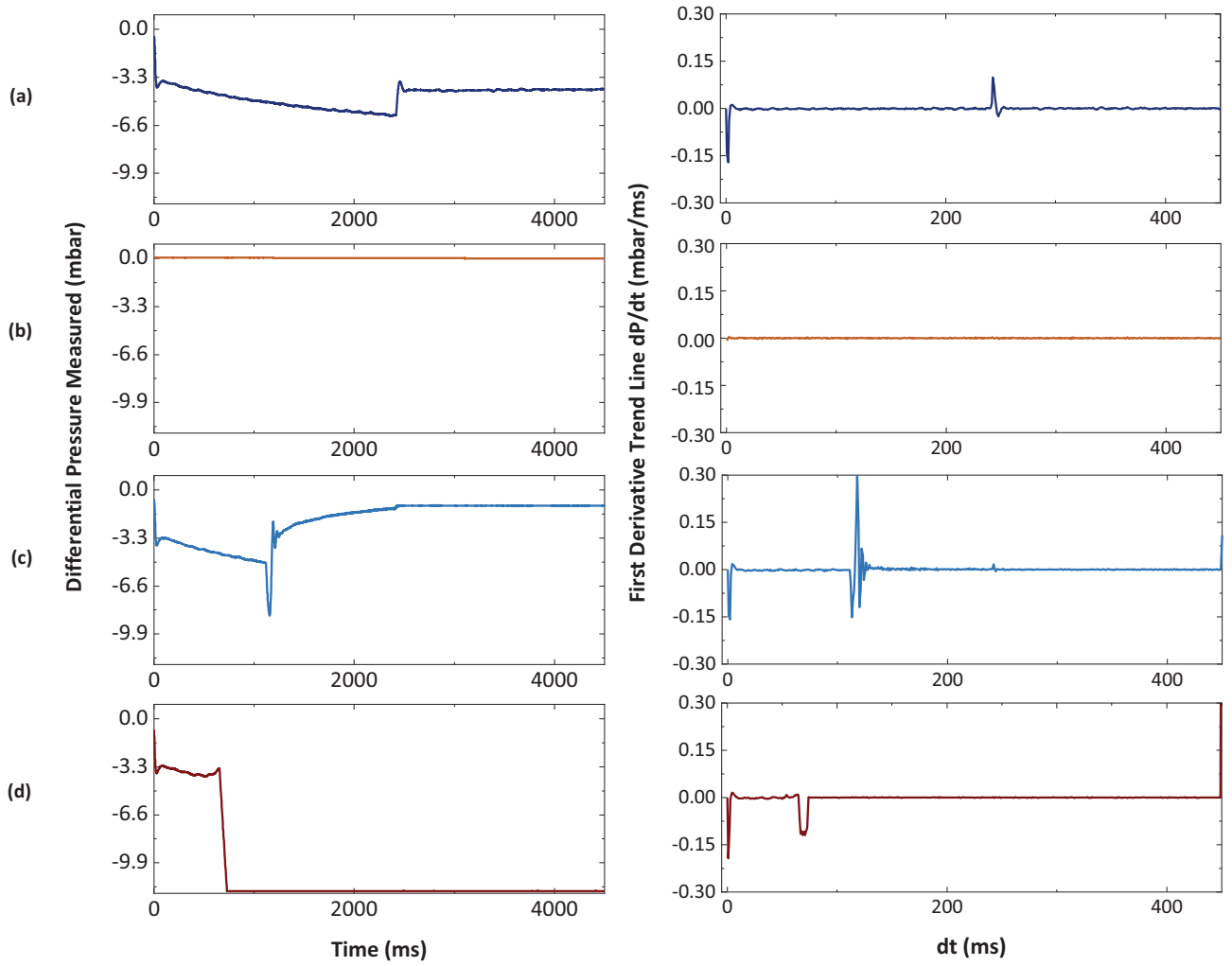


Fig.4. Pressure curves and corresponding first derivative trend lines. (a) Normal aspiration of 200 μ L distilled water with trend line having one sharp valley and peak. (b) No sample aspirated condition wherein the differential pressure measured remains at zero level as tip is exposed to atmosphere. (c) Presence of air bubbles or insufficient liquid in source well resulting in more than one valley and one peak in the first derivative line. (d) Clogging of the pipette tip. As the tip is clogged, differential pressure reaches a saturation as a result of which no peaks can be observed in the trend line.

4) Measurement of liquid volumes aspirated into the tip

Although aforementioned checks determine if the liquid samples are properly aspirated, they will not be able to evaluate if liquid volumes aspirated are within desired range. This generally occurs when there is a small leak in the tip or when the pipette tip is not tightly attached to the holder, which is commonly encountered in automatic liquid handling applications. Detection of liquid level in the pipette tip could help in determining this error. This can also verify loss of calibration of the pipette, which tends to happen over a period of time. To detect level in the tip, pressure data is acquired for extra 2 seconds after the aspiration process and an average of these values gives the constant pressure in the air column. Pipette tip is in contact with the liquid level throughout the acquisition process. Main uncertainty factors which can affect the liquid volume measurements based on pressure sensor can be temperature and humidity of the surroundings. The pressure sensor used is temperature compensated for the range of 0° C to 50° C. An equation has been derived depicting relation between volumes aspirated and the pressure measured for the

water samples, which is explained in Section IV.

Once all these conditions are validated and no abnormalities are detected, operator is updated on the successful aspiration and the algorithm can be immediately switched to dispensing mode.

B. Dispensing Mode

The dispensing algorithm is designed to have an auto-corrective control once the abnormality is detected. Pressure measured after ideal dispensing reaches a zero value as pipette tip would then be exposed to the atmosphere. Pressure curve for normal dispensing is indicated in Fig.5 (a). To dispense liquid samples, pipette takes around 0.1 sec for 10 μ L to 2.5 sec for 200 μ L. Here as well, pressure data is acquired for 2 sec after the dispensing. Average of these values is calculated and is verified to be in the range of ± 0.1 mbar for ideal dispensing. Improper dispensing generally involves liquid droplets hanging on the pipette tip, which result in a non-zero pressure level attained after dispensing as in Fig.5 (b). Once this condition is detected, some corrective action can be designed to eject droplets hanging. One such method can be

touch tip wherein tip is made to contact the side walls of source well. The complete pipetting algorithm is depicted in Fig. 6.

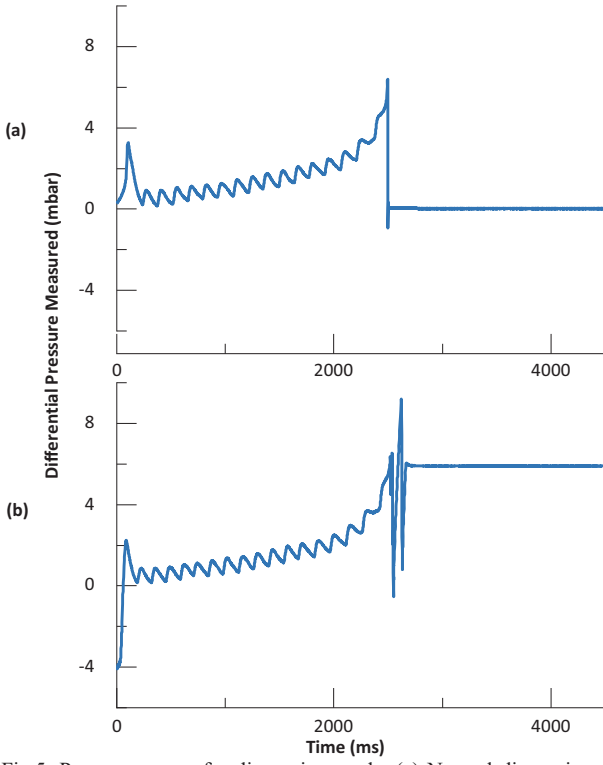


Fig.5. Pressure curves for dispensing mode. (a) Normal dispensing of 200 uL distilled water wherein pressure level reaches a zero value after dispensing. (b) Improper dispensing due to droplet hanging with a non-zero final pressure level.

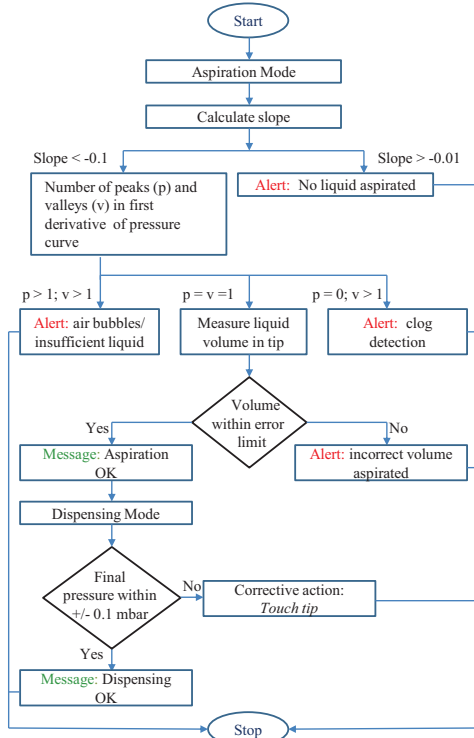


Fig.6. Flow chart of pressure data acquisition algorithm

IV. EXPERIMENTAL STUDY

A. Determination of Repeatability and Accuracy of Pipette

Gravimetric measurements are carried out to determine accuracy and precision of the designed pipetting system. Distilled water which has a density of 1000 kg/m^3 is used for the entire experimental study. Disposable 200uL pipette tips were used with a new tip for each aspiration. Here, mass of the dispensed liquid samples is determined and volume of the samples can be calculated from measured values of mass as depicted in (1)

$$\text{Volume (in uL)} = \text{Mass Measured (in mg)} \times Z \quad (1)$$

Where Z is the conversion factor equal to 1.0038 for distilled water at 1004 mbar atmospheric pressure and 24°C room temperature. An analytical balance of 0.1mg readability (Sartorius, Germany) is used for measurement of the mass of liquid samples.

Repeatability percentage coefficient of variation (%CV) was determined by taking ten repeated measurements for five different dispensed volumes – 10 uL, 50 uL, 100 uL, 150 uL and 200 uL, which gives a measure of imprecision of the system. %CV is calculated by obtaining standard deviation (SD) and mean (M) values for mentioned volumes as indicated in (2)

$$\%CV = \frac{SD}{M} \times 100 \quad (2)$$

Error percentage given by (3) was determined for mentioned volumes, taking an average of ten readings for each volume.

$$\%Error = \frac{MV - EV}{EV} \times 100 \quad (3)$$

Where MV is the measured volume of liquid sample and EV is the expected volume.

B. Measurement of Liquid Level in Pipette Tip

Once liquid sample is aspirated into the tip, pressure in the air column remains constant and is dependent upon pressure exerted by the liquid column, which in turn is dependent on the height of the liquid column in the pipette tip and specific gravity of the liquid sample [7].

Experiments were carried out using distilled water and a mathematical relation between volume aspirated and measured differential pressure is determined. Pressure readings are taken by aspirating water samples from 10 to 200 uL, in steps of 10 uL using a standard 200 uL disposable pipette tip. An average of three readings was taken for each volume for determining the trend line. Pipette tip is in contact with water level throughout the experiments.

V. RESULTS AND DISCUSSIONS

A. Analysis on Accuracy and Repeatability of the System

After performing suitable calibration on the pipetting system, a graph was plotted between measured liquid volumes and the expected volumes as depicted in Fig.7. It can be

observed that the system provides an almost linear output, with measured volumes almost in line with the ideal values. An average non-linearity error of 0.2% was observed. Experimental results obtained for the error percentage and repeatability coefficient of variation are depicted in Table I. It was found that the average error percentage was less than 1.6 while the %CV was below 5.5 throughout the volume range, with as less as 0.3 at 200 uL. These values are observed to be improved over the mean inter-individual imprecision values of manual pipette mentioned, which has %CV of 8.1 at 10uL and 1.1 at 100 uL [4]. These can be further improved when integrated with robotic positioning systems, completely eliminating human intervention for pipetting process.

B. Measurement of Liquid Level in Tip

For measurement of liquid volumes aspirated into the pipette tip, a graph as shown in Fig. 8 was plotted between volumes expected to be aspirated against the differential pressure values obtained, establishing a standard relation for the designed pipette as denoted in (4)

$$y=18.22x^2+34.63x+22.11 \quad (4)$$

For sensitivity analysis of pressure based liquid level measurement, the non-linear response of pressure sensor output with aspirated volumes is divided into three piece-wise linear regions, with input ranges of 5-30 uL, 30-70 uL and 70-200 uL. Corresponding sensitivities are observed to be -0.038 mbar/ uL, -0.017 mbar/ uL and -0.01 mbar/ uL respectively. Rated sensitivity of pressure sensor used being ± 0.02 mbar, minimum volume change required for change in pressure readings is determined to be 0.5 uL, 1.2 uL and 2 uL in the respective linear regions mentioned.

Each time a liquid sample is aspirated, the volume of the same can be calculated and validated if within the permissible error limits and thereby alert operator on the same. A feedback

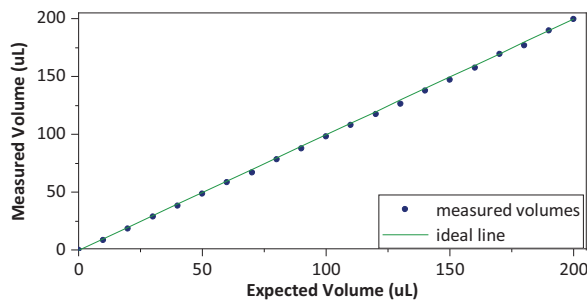


Fig.7. Measured Volume Vs Expected Volume. Line depicts the ideal output trend, points indicate the measured volumes.

TABLE I. EXPERIMENTAL RESULTS FOR ERROR AND % COEFFICIENT OF VARIATION

Volume(uL)	Mean Volume(uL)	% Error	Repeatability %CV
10	9.9880	-0.12	5.5
50	50.6215	1.24	1.5
100	101.6046	1.60	0.5
150	151.8147	1.21	0.4
200	203.1390	1.57	0.3

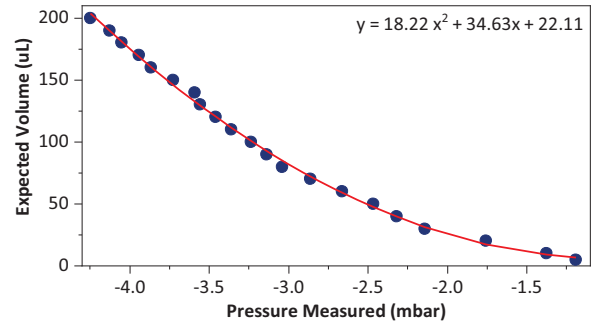


Fig.8. Expected volume to be aspirated Vs Pressure

control can also be devised to automatically compensate for the errors by corresponding piston movement. Check for the liquid volume, along with examining for the mentioned common pipetting errors provide a highly reliable and efficient solution for precise liquid handling in automation of laboratories.

VI. CONCLUSION

This intelligent system forms a completely automated variant of handheld pipetting systems, providing a continuous feedback check to the user. This being able to measure the volume of liquid samples aspirated, provides a self-validating mechanism improving performance accuracy of the system. The obtained values for repeatability and accuracy are observed to be good, with %CV less than 5.5 and error percentage less than 1.6. From the pressure equation derived, volumes of the liquid samples aspirated are calculated for different liquid volumes. These are observed to be within an error band of $\pm 5\%$ from the actual measured volumes and can be improved further. When incorporated with robotic positioning systems, this forms an affordable solution to life science, medical diagnostics and pharmaceutical laboratories, with an increased level of automation, targeting precise handling of liquid samples and minimizing undesirable human errors. Such affordable systems have a large-scale impact on automation of moderately funded laboratories.

REFERENCES

- [1] F. Kong, L. Yuan, Y. F. Zheng, and W. Chen, "Automatic Liquid Handling for Life Science," *J. Lab. Autom.*, vol. 17, no. 3, pp. 169–185, 2012.
- [2] M. Zhang, R. Felder, E. Kim, B. Nelson, and B. Pruitt, "Editorial: Special Issue on Life Science Automation," *IEEE Trans. Autom. Sci. Eng.*, vol. 3, no. 2, pp. 137–140, 2006.
- [3] S. B. Bammesberger, I. Malki, A. Ernst, R. Zengerle, and P. Koltay, "A Calibration-Free, Noncontact, Disposable Liquid Dispensing Cartridge Featuring an Online Process Control," *J. Lab. Autom.*, vol. 19, no. 4, pp. 394–402, 2014.
- [4] G. Lippi, G. Lima-Oliveira, G. Brocco, A. Bassi, and G. L. Salvagno, "Estimating the intra- and inter-individual imprecision of manual pipetting," *Clin. Chem. Lab. Med.*, vol. 55, no. 7, pp. 962–966, 2017.
- [5] M. Berg, K. Undisz, R. Thiericke, P. Zimmermann, and T. Moore, "Evaluation of liquid handling conditions in microplates," *J. Biomol. Screening.*, vol. 6, no. 1, pp. 47–56, 2001.
- [6] T. Sanderson, "PlasmoTron: an open-source platform for automated culture of malaria parasites," *bioRxiv*, pp. 1–5, 2017.
- [7] A.W.Thomas, "Small Volume Pipetting," *J. Assoc. Lab. Autom.*, vol. 3, no. 3, pp. 62–64, 1998.